

RAHUL MITRA

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EDUCATION

National University of Singapore

Singapore

Bachelor of Engineering, Industrial Systems Engineering

Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277): top student in a class of 24, NUS School of Computing.
- Relevant coursework: Simulation, Stochastic Processes, Statistics for Engineering Applications, Operations Research, Artificial Intelligence, Computer Graphics, 3D Computer Vision, Software Engineering, Data Structures and Algorithms.

EXPERIENCE

STMicroelectronics

Singapore

Operations Research (NUS System Design Project), Contract

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Singapore

Robotics Vision Engineer (BlendED AI+X)

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Abbott Laboratories

Singapore

Operational AI Systems & Data Engineer (Contract)

Jun 2026 – Present

- Reverse-engineer four plants' APC steady-state optimizers from their Excel master files into a single HiGHS linear program with a tracking QP and PLC predictor, returning setpoints, binding constraints, and shadow prices at millisecond latency without a commercial solver license.

Abbott Laboratories

Singapore

Data Analytics and Project/Product Management Intern

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling; compared heuristic, MIP, and genetic-algorithm approaches and researched robust optimization.
- Validated a non-destructive, 15-stage changeover-data pipeline against five years of unseen raw data with zero execution failures, normalizing operator- and machine-generated anomalies.

People's Association

Singapore

Platform & Solutions Engineer (Sparks Citizen Developer)

Sep 2025 – Sep 2026

- Delivered a digital garden-plot balloting platform from admin and public frontends through backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.

Republic of Singapore Navy

Singapore

Base Support Assistant

Jan 2022 – Jan 2023

- Compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process with an Excel VBA automation workflow inside a security-constrained environment.

PROJECTS

Volt Pulse SG: SMU Hack For Cities 2026 (Top 8 Finalist of 50+ teams)

Jan 2026

- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

Maritime Deficiency Severity Forecasting: Maritime Hackathon 2025 (Team Lead)

2025

- Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer.

SmartExam: GovTech x NTUPC Product Hackathon

2025

- Built a multi-agent autonomous exam-generation system with a RAG pipeline and Next.js frontend, automating source ingestion, question generation, and education workflows.

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre

Singapore

Elderly Care Volunteer

Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

SKILLS AND CERTIFICATIONS

Technical: CP-SAT, HiGHS linear and quadratic programming, SciPy, hybrid flow-shop scheduling, SimPy, digital-twin simulation, robust optimization, graph optimization, network flow, Dijkstra/SSSP/APSP, decision analytics, statistics, Python, Excel VBA

Certifications: Excel 2019 Advanced (Intelleyz); NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)